Formal Lecture Scribe

Tutorial_1_Solution (1).pdf
PDF

scribe-template (1).pdf
PDF

Using the structure shown in the scribe template above, generate a formal lecture scribe for the tutorial_1_Solution(1) . You must prioritize correctness and include the explicit reasoning provided in the tutorial for each problem from Q1 to Q10. Follow the tutorial's structure and scope exactly, including definitions of distributions like Poisson and Gaussian, specific multinomial formulas, and every step-by-step calculation for the bus arrival, employee attendance, and communication system problems. Ensure all notations and Q-function expressions are faithful to the source material. Do not invent content, introduce new examples, or add intuition not discussed in the provided text.

Below is a **formal lecture scribe** following a typical academic scribe structure (title → definitions → problem statements → derivations). All formulas, notation, and reasoning strictly follow the tutorial solution content without adding extra intuition or new examples. Citations reference the uploaded tutorial.

CSE 400: Fundamentals of Probability in Computing

Tutorial 1 — Formal Lecture Scribe

School of Engineering and Applied Science, Ahmedabad University

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1. Introduction

This tutorial covers several foundational topics in probability theory relevant to computing systems, including:

- Multinomial counting
- Uniform probability models
- Conditional probability
- Poisson distribution
- Gaussian distribution and Q-function
- Binomial reliability models
- Discrete distribution functions

Each problem illustrates a specific probabilistic model and demonstrates the derivation of the required probability expressions step-by-step.

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2. Multinomial Counting Problem (Food Festival)

Problem Statement

Twenty distinct dishes are divided into four platters of five dishes each. The order of dishes within platters and the order of platters themselves are irrelevant.

(a) Number of possible divisions

(b) Probability that each platter contains dishes from a single cuisine, given that each cuisine has exactly five dishes.

Solution

(a) Total Number of Arrangements

There are **20 distinct dishes** partitioned into **4 groups of 5 dishes each**.

Since:

- Order within a group does not matter
- Groups themselves are unlabeled

we use the **multinomial counting formula**.

$$\text{Total ways} = \frac{20!}{(5!)^4 4!}$$

(b) Probability of Cuisine-Pure Platters

Each cuisine contains exactly **five dishes**.

Thus, the only way every platter contains a single cuisine is if **each cuisine forms exactly one platter**.

Because platters are unlabeled, there is **exactly one favorable arrangement**.

Therefore

$$P = \frac{1}{\frac{20!}{(5!)^4 4!}}$$

which simplifies to

$$P = \frac{(5!)^4 4!}{20!}$$

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3. Uniform Distribution Problem (Bus Arrival)

Problem Statement

Buses arrive every **15 minutes starting at 7:00 AM**. A passenger arrives uniformly at random between **7:00 and 7:30**.

Find the probability that the passenger waits:

1. Less than 5 minutes
2. More than 10 minutes

Solution

The arrival time is **uniformly distributed over a 30-minute interval**.

(a) Waiting Time < 5 Minutes

The passenger must arrive within the **final 5 minutes before a bus arrives**.

Possible intervals:

- 7:10–7:15
- 7:25–7:30

Total favorable time:

$$5 + 5 = 10 \text{ minutes}$$

Total sample space:

$$30 \text{ minutes}$$

Therefore

$$P(\text{wait} < 5) = \frac{10}{30} = \frac{1}{3}$$

(b) Waiting Time > 10 Minutes

This occurs if the passenger arrives **more than 10 minutes before the next bus**.

Intervals:

- 7:00–7:05
- 7:15–7:20

Total favorable time:

$$10 \text{ minutes}$$

Thus

$$P(\text{wait} > 10) = \frac{10}{30} = \frac{1}{3}$$

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4. Conditional Probability (Employee Attendance)

Problem Statement

Let

- L = employee arrives late
- E = employee leaves early

Given:

$$P(L) = 0.15, \quad P(E) = 0.25, \quad P(L \cap E) = 0.08$$

Find the probability that the employee **arrives early given that they leave late**.

Solution

Arriving early is the complement of arriving late:

$$L^c$$

Leaving late is the complement of leaving early:

$$E^c$$

First compute

$$P(E^c) = 1 - P(E) = 0.75$$

Next compute

$$P(L \cup E) = P(L) + P(E) - P(L \cap E)$$

$$= 0.15 + 0.25 - 0.08$$

$$= 0.32$$

Using **De Morgan's Law**

$$P(L^c \cap E^c) = 1 - P(L \cup E)$$

$$= 1 - 0.32 = 0.68$$

Therefore

$$P(L^c | E^c) = \frac{P(L^c \cap E^c)}{P(E^c)}$$

$$= \frac{0.68}{0.75}$$

$$\approx 0.907$$

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5. Poisson Distribution (Typographical Errors)

Definition: Poisson Distribution

A random variable X following a Poisson distribution with mean λ has probability mass function

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

Problem

Expected errors per page:

$$\lambda = 0.2$$

Find:

1. $P(X = 0)$
2. $P(X \geq 2)$

(a) Probability of Zero Errors

$$P(X = 0) = \frac{e^{-0.2}(0.2)^0}{0!}$$

$$P(X = 0) = e^{-0.2}$$

$$P(X = 0) = 0.8187$$

(b) Probability of Two or More Errors

Using the complement rule:

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1)$$

$$P(X = 1) = \frac{e^{-0.2}(0.2)^1}{1!}$$

$$= 0.2e^{-0.2}$$

$$= 0.1637$$

Thus

$$P(X \geq 2) = 1 - 0.8187 - 0.1637$$

$$= 0.0176 \approx 0.0175$$

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6. Poisson Distribution (Airplane Crashes)

Let

$$X \sim \text{Poisson}(3.5)$$

(a) At Least Two Accidents

$$P(X \geq 2) = 1 - P(0) - P(1)$$

$$P(0) = e^{-3.5} = 0.0302$$

$$P(1) = 3.5e^{-3.5} = 0.1057$$

Thus

$$P(X \geq 2) = 1 - 0.0302 - 0.1057$$

$$= 0.8641$$

(b) At Most One Accident

$$P(X \leq 1) = P(0) + P(1)$$

$$= 0.0302 + 0.1057$$

$$= 0.1359$$

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7. Poisson Distribution (Highway Accidents)

Let

$$X \sim Poisson(3)$$

(a) Probability $X \geq 3$

$$P(X \geq 3) = 1 - P(0) - P(1) - P(2)$$

$$P(0) = e^{-3} = 0.0498$$

$$P(1) = 3e^{-3} = 0.1494$$

$$P(2) = \frac{3^2}{2!}e^{-3} = 0.2240$$

Thus

$$P(X \geq 3) = 1 - 0.4232$$

$$= 0.5768$$

(b) Conditional Probability

$$P(X \geq 3 | X \geq 1) = \frac{P(X \geq 3)}{P(X \geq 1)}$$

$$P(X \geq 1) = 1 - P(0) = 0.9502$$

Thus

$$P(X \geq 3 | X \geq 1) = \frac{0.5768}{0.9502}$$

$$\approx 0.607$$

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8. Gaussian Distribution and Q-Function

Given PDF

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Comparing with the Gaussian form

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

we obtain

$$m = -3, \quad \sigma^2 = 4, \quad \sigma = 2$$

Relations:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$P(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

Results

1.

$$P(X \leq 0) = \Phi\left(\frac{3}{2}\right) = 1 - Q\left(\frac{3}{2}\right)$$

2.

$$P(X > 4) = Q\left(\frac{7}{2}\right)$$

3.

$$|X+3| < 2 \Rightarrow -5 < X < -1$$

$$P = \Phi(1) - \Phi(-1)$$

$$= 2\Phi(1) - 1$$

$$= 1 - 2Q(1)$$

4.

$$|X - 2| > 1$$

$$P = F_X(1) + P(X > 3)$$

$$= 1 - Q(2) + Q(3)$$

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9. Jury Decision Problem

Let

- θ = probability a juror makes the correct decision
- α = probability defendant is guilty

If the defendant is **innocent**, a correct verdict requires **at most 7 guilty votes**.

$$\sum_{i=5}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

If the defendant is **guilty**, a correct verdict requires **at least 8 guilty votes**.

$$\sum_{i=8}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

Therefore the total probability of a correct decision is

$$\alpha \sum_{i=8}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i} + (1-\alpha) \sum_{i=5}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

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10. Poisson PMF Identification

Given

$$p(i) = c\frac{\lambda^i}{i!}$$

Using normalization

$$\sum_{i=0}^{\infty} p(i) = 1$$

$$c\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Since

$$e^x = \sum_{i=0}^{\infty} \frac{x^i}{i!}$$

we obtain

$$ce^{\lambda} = 1$$

$$c = e^{-\lambda}$$

Results

$$P(X = 0) = e^{-\lambda}$$

$$P(X > 2) = 1 - e^{-\lambda} - \lambda e^{-\lambda} - \frac{\lambda^2 e^{-\lambda}}{2}$$

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11. Reliability of Communication Systems

Model

Each component functions independently with probability p.

The system works if **at least half of the components function**.

(a) Comparison of 5-Component and 3-Component Systems

For 5 components:

$$\binom{5}{3} p^3 (1-p)^2 + \binom{5}{4} p^4 (1-p) + p^5$$

For 3 components:

$$\binom{3}{2}p^2(1-p)+p^3$$

The inequality simplifies to

$$3(p-1)^2(2p-1)>0$$

Thus

$$p>\frac{1}{2}$$

(b) General Result

For systems with $2k+1$ and $2k-1$ components, the difference in success probabilities is

$$\binom{2k-1}{k}p^k(1-p)^k(2p-1)$$

Since the coefficient is positive,

$$P_{2k+1}>P_{2k-1}$$

iff

$$p>\frac{1}{2}$$

If you want, I can also **convert this entire scribe into a clean LaTeX document (Overleaf-ready)** with sections, theorem boxes, and aligned equations exactly like a typical **CSE probability course scribe submission**.